Supervised Learning with Tabular Health Data

Predict which patients will come back to hospital, and decide what a wrong guess costs

Supervised learning | A table of patient records | Uses Weeks 1–4 and 7

The task. When a patient leaves hospital, doctors would like to know: how likely is this person to be back within 30 days? You will build a model that answers this from patient records. You will also decide how the hospital should use it, and check whether it works equally well for different groups of patients.

THE DATA

In Week 9, the teaching team gives you a table of patient records with names and IDs removed. It has a few thousand patients and a few dozen columns, such as age, test results, diagnoses, medicines and past hospital stays. Some values are missing, and most patients do not come back, so the two outcomes are unbalanced. The data is not public, so no AI assistant has seen it. Health data is legally sensitive: keep it on your own computer and never paste it into an online tool.

WHAT YOU WILL DO

1. Explore the data and prepare it carefully, so that nothing recorded after the patient left sneaks in (Week 1).
2. Train simple models first: one that always says "will not come back", logistic regression, and a tree-based model. Plot their learning curves (Week 2).
3. Build your own neural network for the same task, and plot its loss and accuracy as it trains (Week 7).
4. Choose the cut-off for flagging a patient by weighing the two kinds of mistake: missing a patient who comes back, or calling one who did not need it. Report how many returning patients you catch, how many flags are right, and whether the model's percentages can be trusted.

WHAT YOU CODE YOURSELF

You write three things yourself in NumPy and test each one against a library: the network's **back-propagation** (checked against PyTorch and by nudging), its **optimiser step**, and **PCA via SVD** (checked against scikit-learn), which you use to explore the patient table.

IS IT FAIR AND SAFE?

Check whether the model catches returning patients equally well for men and women and for different age groups. Try one fix, such as a different cut-off for each group, and measure what it costs. Use SHAP to show which columns drive the predictions, and check that none is secretly standing in for a group. Write a model card and a short note on the legal side: health data under GDPR, and why the EU AI Act treats a tool like this as high-risk.

QUESTIONS TO PREPARE FOR

Why did you choose that cut-off, and who pays for each kind of mistake? When your model says 70%, do 70% of those patients really come back? Which column would worry a doctor, and why?

COMMON MISTAKES

Columns that are only filled in after the patient leaves, which give the answer away. The same patient appearing in both your training and your test data. A high accuracy that only means most patients do not come back.

WEEK-BY-WEEK PLAN

Week 9: data explored and prepared; simple models trained, with learning curves. **Week 10:** your network trained; your three hand-written pieces passing their tests. **Week 11:** cut-off chosen; percentages checked; fairness check with one fix; SHAP; model card. **By 6 January 2027:** report and repository submitted. **Week of 11 January:** 10-minute in-person interview.

TO AIM HIGHER

Show how the best cut-off changes as a missed patient becomes more costly. If the model's percentages are off, fix them and show the improvement on a plot. Put error bars on the differences between groups, so you know which ones are real.